BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI, K. K. BIRLA GOA CAMPUS
II SEMESTER 2025-2026
COURSE HANDOUT (PART II)

Date: 06/01/2026

In addition to Part I (the General Handout for all courses, appended to the timetable), this section provides further specific details regarding the course.

Course No.	**:**	**CS F212**
Course Title	**:**	**DATABASE SYSTEMS**
Instructor-in-charge	**:**	**Dr. Shubhangi K. Gawali**
Lab Instructors	**:**	**Dr. Baskar** **abaskar@goa.bits-pilani.ac.in** **Dr. Shubhangi Gawali** **shubhangi@goa.bits-pilani.ac.in, Chamber No. D-263**

1. Scope and Objective of the course

The scope of the course encompasses database system concepts and major application areas. The objective is to understand various data models and to develop the relational model of the database, including the rigorous practice of query language, SQL. The emphasis is to apply the concepts to a wide range of applications.

2. Text Book:

T1 Raghu Ramakrishnan and Johannes Gehrke, **Database Management System**, 3rd Ed, McGraw-Hill, 2002

Reference Books:

R1 Elmasri, Navathe, **Fundamentals of Database Systems**, Pearson Education, 2002
R2 Silberschatz A, Korth H F, and Sudarshan S, **Database System Concepts**, TMH, 2002

3. Lecture Plan

Lectures	Topic	Reference
1-3	Overview of Database Systems, File System versus Database System, Structure of Database System and its users, Database Applications	T1: Ch 1
4-6	Introduction to database design, ER model	T1: Ch 2
7-9	Enhanced E-R model	R1: Ch 4
10-14	Relational model concepts, ER to Relational model	T1: Ch 3
15-17	The relational algebra and extended relational algebra operations, Formation of queries, Modification of the database, and views.	T1: Ch 4
18-21	SQL: Queries and Constraints	T1: Ch 5
22-23	Functions, Procedures and Triggers	Class notes
24-25	Integrity and Security: Domain Constraints, Referential integrity, Triggers in SQL, Security and authorization in SQL	T1: Ch 21
26-29	Relational-Database Design:	T1: Ch 19

	Pitfalls in relational database design, Functional dependencies, Decomposition, normal forms, and Normalization techniques.	
30-33	Transaction Management	T1: Ch 16
34-36	Concurrency Control	T1: Ch 17
37-38	Crash Recovery	T1: Ch 18
39-40	Storage and indexing	T1: Ch 8,9

4. Evaluative labs schedule

Month	Thursdays date	Lab No	Topic
Jan	15	1	DDL and DML commands
	29	2	aggregate functions, order by, pattern search
Feb	5	3	Mapping of ER diagram to Relational Model and Constraints (check, Ref integrity etc)
	19	4	Groupby and having clause and Set operations
	26	5	Joins and Nested queries
Mar	5	6	procedure and user defined functions
	19	7	views, triggers
Apr	9	8	EER to RM

5. Project Evaluation dates:

Month	Thursdays Date	plan
Jan	15	Project title and team members name
Jan	29	Project ER diagram, tools and and role of each member submission
Feb	12	Project eval
Apr	2	Project eval
Apr	16	Project eval

6. Evaluation Scheme:

	Theory		Practicals	Group project	Quizzes
	Mid-semester	**Comprehensive**	**Best 5 out of n eval labs, 7>n>5**	**Application development**	**Pen paper based**
Date	13th Mar 2026	13th May 2026	Thursdays, according to the lab plan	As per the plan, last eval on 16th Apr Thursday	Class hours
Time	2 to 3:30 pm	2 to 5 pm	11 to 1 pm	Will be announced	Class hours
Weightage (%)	30%	35%	25%	5%	5%
Nature of Component	Closed Book	Closed Book	Open Book (limited)	Open Book	Open Book

7. Malpractice Regulations:

1. Any student or team of students found involved in malpractices in any examination component will be awarded negative marks equal to the weightage of that component, and all previous evaluative components will be made zero. The student(s) will be blacklisted.
2. Any student or team of students found repeatedly, more than once across all courses, involved in malpractices will be reported to the Disciplinary Committee for further action. This will be in addition to the sanction mentioned above.
3. A mal-practice - in this context - will include but not be limited to:
   - Submitting some other student's / team's solution(s) as one's own;
   - Copying some other student's / team's data or code or other forms of a solution;
   - Seeing some other student's / team's data or code or other forms of a solution;
   - Permitting some other student / team to see or to copy or to submit one's own solution;
   - OR other equivalent forms of plagiarism wherein the student or team does not work out the solution and/or uses some other solution or part thereof (such as downloading it from the web).
4. The degree of mal-practice (the size of the solution involved or the number of students involved) will not be considered as mitigating evidence. Failure on the part of instructor(s) to detect mal-practice at or before the time of evaluation may not prevent sanctions later on.
5. The decision of punishment at course-level will be taken by Instructor-InCharge.

5. **Chamber Consultation Hour**: Thursday 12 noon to 1 pm. with prior appointments through email.

6. **Notice**: Notice concerning this course will be displayed on Quantaaws notice board.

7. Makeup Policy:

- Permission of the Instructor-in-Charge is required to take a make-up
- Make-up applications must be given to the Instructor-in-charge personally.
- A make-up test shall be granted only in genuine cases where - in the Instructor's judgment - the student would be physically unable to appear for the test.
- In case of an unanticipated illness preventing a student from appearing for a test, the student must present a Medical Certificate from BITS medical centre.
- Requests for make-up for the comprehensive examination – under any circumstances – can only be made to In-charge, Instruction Division.

8. Criterion for NC: 30% of the average of the top three performers or 40% of median, whichever is lower, or a suitable modification at the discretion of the IC with the concurrence of the Department.

Instructor-In-Charge
CS F212